NLD - Non-Linear Dynamic Chaos

Signature Sheet

Student's Name _____ Partner's Name _____

Pre-Lab Discussion Questions

It is your responsibility to discuss this lab with an instructor before your first day of your scheduled lab period. This signed sheet must be included as the first page of your report. Without it you will lose grade points. You should be prepared to discuss at least the following before you come to lab:

1. What is Non-linear Dynamics (NLD)? Specifically what is it that is non-linear?
2. What is chaos? What are the defining characteristics of chaos?
3. Present and solve on the board the problems 9.3.1 and 11.4.2 in Ref. 1.
4. What is the Feigenbaum ratio? What does it mean? See the Wolfram Math World subjects for help: Feigenbaum and Chaos.
5. What is a return map? What is a Poincaré map?
6. What is a Fourier Transform? What does the power spectrum of a square wave look like? What is aliasing?

Staff Signature _____ Date _____

Completed before the first day of lab? (Circle one) Yes / No

Mid-Lab Discussion Questions

1. On day 4 of this lab, demonstrate the Cobweb VI that you wrote, showing regions of chaos and periodicity.

Staff Signature _____ Date _____

Completed by day 4 of lab? (Circle one) Yes / No

Checkpoint Signatures

1. DFT Window Units

   Staff Signature _____

2. Cob Web Program

   Staff Signature _____

3. Liapunov Exponent Program

   Staff Signature _____

4. Henon Map

   Staff Signature _____

5. Bouncing Ball and PN-Junction Circuits

   Staff Signature _____